

Alg2 NOTES 7.2 - PROPERTIES OF EXPONENTIAL FUNCTIONS

Parent Graphs: $y = b^x$ New to us! So let's get some points to see its shape.

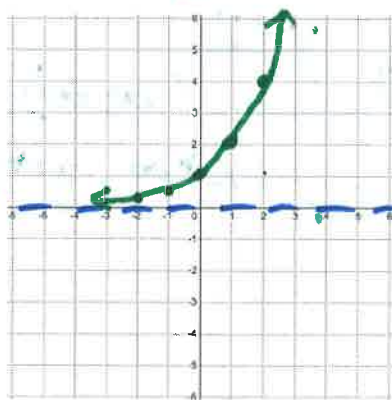
Example:

$y = 2^x$

pick #s around 0

x	y
-2	$2^{-2} = \frac{1}{4}$
-1	$2^{-1} = \frac{1}{2}$
0	$2^0 = 1$
1	$2^1 = 2$
2	$2^2 = 4$

$b > 1$ ex $y = 2^x$



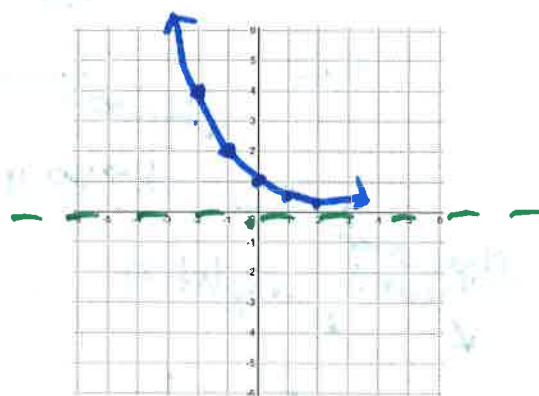
Example:

$b < 1$

$y = (\frac{1}{2})^x$

x	y
-2	$(\frac{1}{2})^{-2} = 4$
-1	$(\frac{1}{2})^{-1} = 2$
0	$(\frac{1}{2})^0 = 1$
1	$(\frac{1}{2})^1 = \frac{1}{2}$

$0 < b < 1$



Get two key points:

When $x = 0$, $(0, 1)$

When $x = 1$, $(1, 2)$

It's Exponential growth

This function increases on its domain

goes up as we read Left to right

Eqtn of Horizontal asymptote: $y = 0$

Domain: $(-\infty, \infty)$ Range: $(0, \infty)$
(interval notation) ↑ smaller larger

End behavior: $x \rightarrow -\infty$, $y \rightarrow 0$

$x \rightarrow \infty$, $y \rightarrow \infty$

Get two key points:

When $x = 0$, $(0, 1)$

When $x = 1$, $(1, \frac{1}{2})$

It's Exponential Decay

This function decreases on its domain

Eqtn of Horizontal asymptote: $y = 0$

Domain: $(-\infty, \infty)$ Range: $(0, \infty)$
(interval notation)

End behavior: $x \rightarrow -\infty$, $y \rightarrow \infty$

$x \rightarrow \infty$, $y \rightarrow 0$

II. TRANSFORMATIONS: $y = a \cdot b^{x \pm c} \pm d$

vert. stretch

Sketch and find the features of each graph.

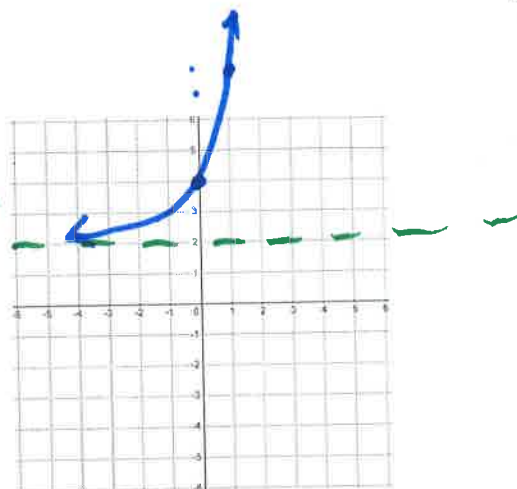
1. $f(x) = 2(3)^x + 2$ ← up 2 → more asymptote first

Use your knowledge of the shape and then find 2 pts. Good points are when the exponent = 0 and the exponent = 1

exp = 0
 $y = 2(3)^0 + 2$
 $= 2(1) + 2$
 $= 4$
 (0, 4)

exp = 1
 $y = 2(3)^1 + 2$
 $y = 6 + 2 = 8$
 (1, 8)

⇒ sketch



The function $f(x)$ is logarithmic exponential with a vertical horizontal asymptote of $y = 2$

The range of the function is $(2, \infty)$ and it is increasing decreasing on its domain of $(-\infty, \infty)$. The

end behavior on the LEFT side is as $x \rightarrow -\infty$, $y \rightarrow 2$, and on the RIGHT side is as $x \rightarrow \infty$, $y \rightarrow \infty$

2. $f(x) = -3(5)^{x-4} + 1$ ← right 4
 Flipped over x
 ↑ up 1
 Likes! shifted up 1

exp = 0 sub in
 $x - 4 = 0$
 $x = 4$

Find y when $x = 4$

$y = -3(5)^{4-4} + 1$

$y = -3(1) + 1 = -2$

(4, -2)

exp = 1

$x - 4 = 1$

$x = 5$

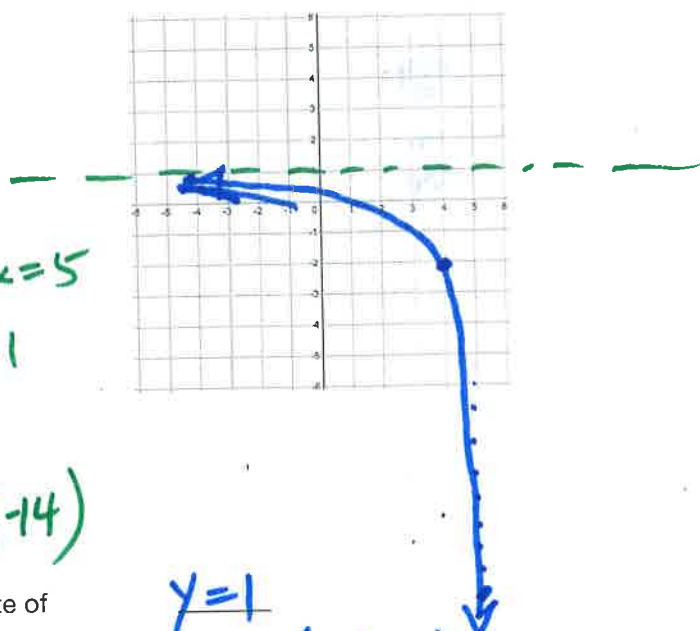
Find y when $x = 5$

$y = -3(5)^{5-4} + 1$

$y = -3(5) + 1$

$y = -14$

(5, -14)



The function $f(x)$ is logarithmic exponential with a vertical horizontal asymptote of $y = 1$

The range of the function is $(-\infty, 1)$ and it is increasing decreasing on its domain of $(-\infty, \infty)$. The

end behavior on the LEFT side is as $x \rightarrow \infty$, $y \rightarrow 2$, and on the RIGHT side is as $x \rightarrow \infty$, $y \rightarrow -\infty$

III. WRITING AN EXPONENTIAL EQUATION GIVEN 2 POINTS

3. Write an exponential function in the form $y = ab^x$ that goes through the points $(0, 7)$ and $(2, 175)$.

↑ general form of exp. eqn.

For specific equation you must find a and b and sub them in

1st Always sub the point that has $x=0$, in first

2nd get a

$$y = ab^x \rightarrow 7 = ab^0 \rightarrow 7 = a(1) \Rightarrow a = 7$$

3 Now sub in your $a \rightarrow y = 7b^x$

4 Now sub in other point $(2, 175)$

$$175 = 7b^2$$

$$25 = b^2$$

$$b = 5$$

(only take positive $\sqrt{\text{root}}$)

6 Write it!

$$y = 7(5)^x$$

4. Write an exponential function in the form $y = ab^x$ that goes through the points $(0, 9)$ and $(3, 243)$.

$$9 = ab^x$$

$$a = 9$$

$$y = 9b^x$$

$$243 = 9b^3$$

$$27 = b^3$$

$$b = 3$$

$$y = 9(3)^x$$